Original Article

Adjuvant therapy with heparin in patients with lung cancer without indication for anticoagulants: A systematic review of the literature with meta-analysis

Yuman Yu, Qun Lv, Bin Zhang[1], Fen Lan[1], Yifan Dai

Department of Pulmonary Medicine, Hangzhou Normal University, The Affiliated Hospital of Hangzhou Normal University, Hangzhou 310015, Zhejiang, [1]Department of Respiratory and Critical Care Medicine, Second Affiliated Hospital, Institute of Respiratory Diseases, Zhejiang University School of Medicine, Hangzhou, China

For correspondence: Dr. Yifan Dai, Department of Pulmonary Medicine, Hangzhou Normal University, The Affiliated Hospital of Hangzhou Normal University, Hangzhou 310015, Zhejiang, China. E-mail: jiang1870194@yeah.net

ABSTRACT

Background: The effect of heparin in improving cancer survival has gained increasing attention over the past decades. Several clinical trials have evaluated the role of heparin on survival outcome and its safety profile in lung cancer patients. Thus, we performed a systematic review and meta-analysis from the results of randomized controlled trials (RCTs) to assess the efficacy and safety of heparin in patients with lung cancer without indication for anticoagulants.

Methods: We searched PubMed, Embase, and The Cochrane Central Register of Controlled Trials databases for relevant studies. The inclusion criteria used were patients with lung cancer without a concurrent diagnosis of venous thromboembolism (VTE) and were treated with low-molecular-weight heparin (LMWH) or unfractionated heparin (UFH). The outcomes included survival outcome, VTE, bleeding, major bleeding, and thrombocytopenia. The results were presented as hazard ratio (HR) and relative risk (RR), and the STATA 12.0 package was used for comprehensive quantitative analysis.

Results: A total of 6 studies with 753 cases and 640 controls were included for the final analysis. The meta-analysis showed significant differences in survival with an HR of 0.71 (95% confidence interval [CI] 0.60–0.84), particularly in limited-stage small cell lung cancer (SCLC) with an HR of 0.57 (95% CI 0.43–0.77), and also in VTE (RR 0.46; 95% CI 0.27–0.80) when heparin was compared with placebo or no anticoagulant. There were no significant differences in risks for bleeding (RR 1.53; 95% CI 0.96–2.45), major bleeding (RR 1.43; 95% CI 0.59–3.45), and thrombocytopenia (RR 0.86; 95% CI 0.66–1.12).

Conclusion: Administration of heparin (mainly LMWH) as primary thromboprophylaxis for lung cancer patients without indication for anticoagulants was associated with a significant survival benefit, particularly in limited-stage SCLC.

KEY WORDS: Anticoagulants, heparin, lung cancer, systematic review

INTRODUCTION

Worldwide, lung cancer accounts for approximately 1.6 million new cases of cancer and 1.4 million of cancer-related death annually.[1] Lung cancer is associated with a poor prognosis. For non-small cell lung cancer (NSCLC), which is the most frequent histological type of lung cancer, the overall 5-year survival rate is 18.2%. The prognosis is even poorer in SCLC, which has an overall 5-year survival rate of 6.3%.[2] Long-term survival for lung cancer patients remains a tremendous challenge, and there is an urgent need for new approaches to enhance our current therapeutic approaches.[3]

The association between venous thromboembolism (VTE) and cancer has been convincingly demonstrated since Trousseau's initial observations in 1865.[4] The overall risk of VTE is increased 7-fold in patients with a malignancy than persons without malignancy, and this risk is further increased by chemotherapy, surgery, and disease progression.[5,6] In patients with lung cancer, even a 22-fold increased risk was observed.[5] On the other hand, VTE in cancer patients predicts a poor prognosis.[7,8] The occurrence of cancer-associated thrombosis presents a dilemma for the physician, especially in patients with metastatic disease.[9]

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Cite this article as: Yu Y, Lv Q, Zhang B, Lan F, Dai Y. Adjuvant therapy with heparin in patients with lung cancer without indication for anticoagulants: A systematic review of the literature with meta-analysis. J Can Res Ther 2016;12:37-42.

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Thence preventing VTE in patients with cancer may improve cancer survival.

Due to the importance of thromboprophylaxis in cancer patients, numerous studies have been conducted to explore the impact of anticoagulants on cancer survival. It seems that heparin is superior to warfarin in cancer patients in terms of increased efficacy with a good safety profile and reliability and is also associated with improved quality of life.[10]

Besides its antithrombosis property in cancer, heparin also exhibits a potential anticancer effect. The anticancer mechanism of heparin remains unclear. It has been reported that heparin possesses direct antitumor effects, anti-angiogenic effects, and immunomodulatory effects.[11] Heparin also enhances antitumor effects of chemotherapeutic agents.[12,13]

Several studies have evaluated the role of heparin on survival outcome and safety profile in lung cancer patients. Three small-scale studies have indicated that adjuvant therapy with heparin may be associated with a survival benefit in SCLC patients.[13-15] Nevertheless, a randomized clinical trial examining the effect of nadroparin on survival cancer patients did not show a survival benefit of nadroparin in cancer patients, neither in the NSCLC subgroup.[16] However, most of these trials were too small to reliably detect moderate treatment effects.

Consequently, the survival outcome and safety of heparin in lung cancer patients remains inconclusive owing to the limited sample size and inconsistent results of these individual studies. We, therefore, conduct a systematic literature review and meta-analysis of six randomized controlled trials (RCTs) concerning this subject. The purpose of our study was to determine the efficacy and safety of heparin on survival in patients with lung cancer.

METHODS

Literature search

We performed a comprehensive literature search in the PubMed, Embase, and The Cochrane Central Register of Controlled Trials databases. The terms (tumor OR malign * OR carcinoma * OR cancer OR SCLC OR non-SCLC OR NSCLC OR neoplasm*) AND (heparin OR low molecular weight heparin OR LMWH OR enoxaparin OR dalteparin OR reviparin OR certoparin OR tinzaparin OR bemiparin OR nadroparin OR * parin) were used. We scanned references of all included trials and reviews for additional studies. In addition, we searched databases of ongoing and unpublished trials on line (http://www.clinicaltrials.gov).

Study selection

We included RCTs that compared the addition of LMWH or unfractionated heparin (UFH) to standard chemotherapy in lung cancer patients without a concurrent diagnosis of VTE as primary thromboprophylaxis. To ensure an adequate amount of therapeutic time, LWMH or UFH had to be given for at least 4 weeks without interruption. The primary efficacy outcome was survival outcome. Other outcomes also evaluated were (1) symptomatic DVT, symptomatic pulmonary embolism, and all reported thromboembolic events, (2) adverse effects including major bleeding, minor bleeding, and thrombocytopenia.

Data extraction and study outcomes

Two reviewers (Y. Yu and F. Lan) independently extracted the data on basic study design, patient characteristics, study outcomes, and measures of study quality. A third investigator was consulted in case of disagreement (B. Zhang). Data for analysis were abstracted systematically from the published reports and included: (1) authors and year of publication, (2) the number of patients included in the study, (3) type and stage of lung cancer and other patient characteristics, (4) types, doses, and duration of anticoagulation therapy, (5) clinical outcomes, and (6) study design and Jadad score.

Quality assessment

The methodological quality of each trial was evaluated by using the Jadad score,[17,18] a validated quality assessment instrument which incorporates randomization, blinding, and attrition to derive a score of 0 (very poor) to 5 (rigorous). Two reviewers independently assessed the quality of the included trials. Any divergence was resolved by discussion with a third reviewer to reach an agreement.

Statistical analysis

We measured the survival outcome by hazard ratio (HR) between the survival distributions. The log (HR) and its variance were extracted from the publications. When such data were not available directly, we extracted from Kaplan–Meier survival curves. Survival rates at some specified times were chosen and digitized by Get Data Graph Digitizer. We estimated the log rank statistic and its variance by using Parmar's methods.[19] The individual HR point estimates were combined in STATA 12.0 (http://www.stata.com; Stata Corporation, College Station, TX), to obtain a pool HR estimate of the treatment effect. Conventionally, an HR <1 implied a survival benefit for the treatment arm. For dichotomous data, RRs and 95% confidence intervals (CIs) were calculated separately for each study to assess the incidence of outcomes by experimental group. An RR <1.0 was in favor of heparin therapy. We then pooled the RRs in STATA 12.0. The I^2 statistics was used to quantify the heterogeneity across the studies,[20] with I^2 $<25\%$, 25–75%, and $>75\%$ to represent the low, moderate, and high degree of heterogeneity, respectively. The HRs or RRs were pooled using a random effects model when I^2 was $>50\%$ or a fixed effects model when I^2 was $<50\%$. Publication bias was estimated by the funnel plot for the primary outcome.[21]

RESULTS

We found 9076 articles by search, and finally identified six studies eligible for our meta-analysis. The process of study selection was illustrated in Figure 1.

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Studies characteristics

A total number of 1393 participants were included, 753 patients in the heparin group, and 640 patients in the control group. The intervention was UFH in one study[13] and LMWH in 5 studies.[14-16,22,23] Four studies reported survival outcomes. The details of the study characteristics are described in Table 1. The Begger's funnel plot and Begger's test were conducted to assess the publication bias. Statistical results did not demonstrate publication bias ($P > 0.05$ for all).

Survival

Across all eligible studies, according to a fixed-effects model, heparin had a discernible effect on survival in lung cancer patients with a calculated HR of 0.71 (95% CI 0.60–0.84; $I^2 = 42.0\%$) as described in Figure 2. A further pool estimate of the SCLC subgroup showed significant survival benefit (HR 0.72; 95% CI 0.59–0.87; $I^2 = 47.4\%$). We also performed a subgroup analysis of the limited-stage SCLC group in the fixed-effects model and identified substantially improvement in survival (HR 0.57; 95% CI 0.43–0.77; $I^2 = 0$; Figure 3).

Thromboembolic events

There were 19 thromboembolic events among the 529 patients who received heparin therapy and 34 events in 404 patients in the control group. According to a fixed-effects model, there was a significant difference in thromboembolic events between the two groups (RR 0.46; 95% CI 0.27–0.80; $I^2 = 0\%$; Table 2).

Bleeding events

Based on four studies,[14,15,22,23] bleeding occurred in 42 of 534 (7.9%) patients in the heparin group versus 25 of 413 (6.1%) patients in the control group. In the fixed-effects model, LMWH use did not increase either bleeding (RR 1.53; 95% CI, 0.96–2.45; $I^2 = 6.2\%$; Table 2) or major bleeding complications (RR 1.43; 95% CI, 0.59–3.45; $I^2 = 0\%$; Table 2). Major bleeding was defined as bleeding which was fatal, requiring transfusion of two units of packed red cells, or associated with a decrease in hemoglobin of 20 g/L (e.g., retroperitoneal, intracranial, and hemoptysis).

Thrombocytopenia

In the three studies that reported thrombocytopenia, 78 of 335 (23.3%) patients in heparin group experienced a thrombocytopenia complication, compared with 95 of 332 (28.6%) in control group. According to a fixed-effects model, no significant difference was seen (RR 0.86; 95% CI, 0.66–1.12; $I^2 = 13.7\%$; Table 2).

DISCUSSION

The results of our meta-analysis indicated that administration of heparin (mainly LMWH) as primary thromboprophylaxis for lung cancer patients without indication for anticoagulants is associated with statistically significant survival benefit, particularly in limited-stage SCLC. A significant reduction in thromboembolic events was observed with the heparin therapy while the risk for bleeding, major bleeding, and thrombocytopenia is not significantly increased compared with the control arm.

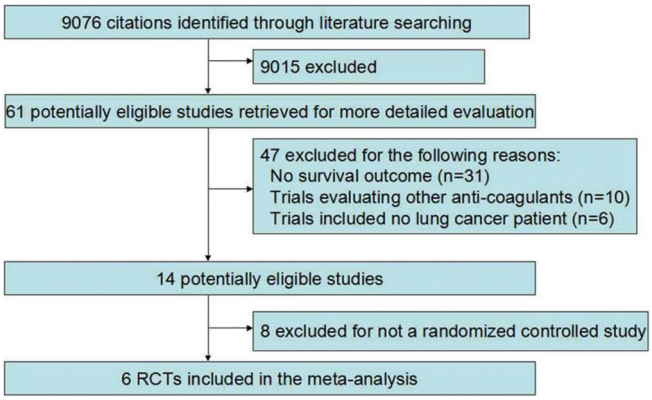

Figure 1: Flow diagram of the systematic review

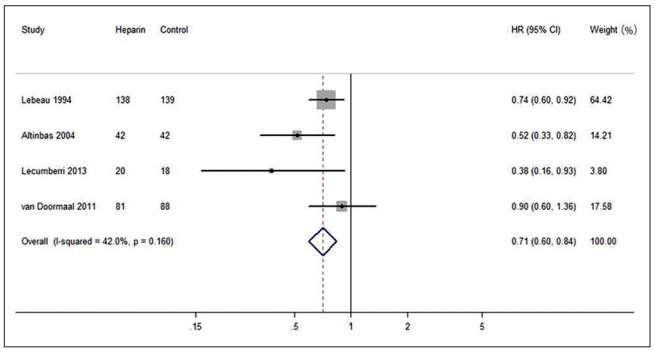

Figure 2: The effect of heparin therapy on survival in patients with lung cancer

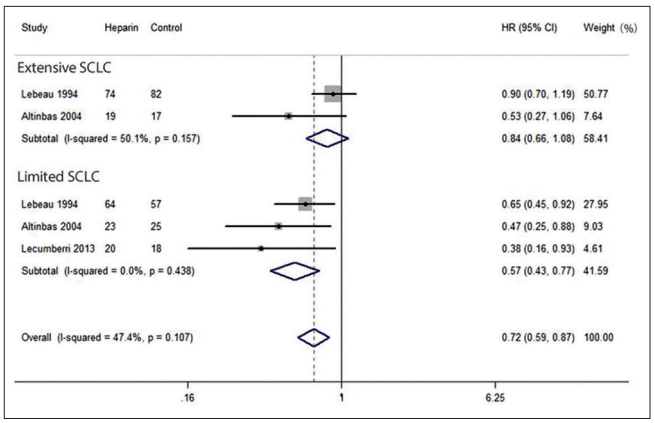

Figure 3: The effect of heparin therapy on survival in patients with small cell lung cancer

So far several meta-analysis studies have investigated the role of heparin played in cancer treatment. Akl *et al*. focused their meta-analysis on 5 five RCTs that compared heparin-treated cancer patients with a control group in 2008 and showed a survival advantage of heparin in cancer patients. They performed a subgroup analysis of SCLC patients in two RCTs

Table 1: Basic characteristics of individual trials

Study	Number of patients	Participants	Interventions	Outcome assessment	Study design	Jadad score
Lebeau *et al*., 1994	138/139	Patients with limited or extensive SCLC undergoing CT	Intervention: Subcutaneous heparin for 5 weeks Control: No intervention	Survival outcome	Open-label randomized controlled trial	3
Altinbas *et al*., 2004	42/42	Patients with SCLC undergoing CT	Intervention: Dalteparin 5000 IU once daily during 18 weeks of CT Control: No intervention	Survival outcome, thromboembolic events, minor bleeding, and thrombocytopenia	Open-label randomized controlled trial	2
Agnelli *et al*., 2009	199/80	Patients with metastatic or locally advanced lung cancer undergoing CT	Intervention: Subcutaneous nadroparin 3800 IU anti-Xa once daily for a maximum of 4 months Control: Placebo	Thromboembolic events, and major bleeding	Double-blind randomized placebo-controlled trial	4
Van Doormaal *et al*., 2011	81/88	Patients with Stage IIIB NSCLC receiving standard anticancer treatment	Intervention: Subcutaneous nadroparin for 6 weeks Control: No intervention	Survival outcome, and major bleeding	Open-label randomized controlled trial	3
Haas *et al*., 2012	273/273	patients with Stage III or IV NSCLC undergoing CT	Intervention: Certoparin 3000 IU subcutaneously once daily for 6 months Control: Placebo	Thromboembolic events, major and minor bleeding, and thrombocytopenia	Double-blind randomized placebo-controlled trial	5
Lecumberri *et al*., 2013	20/18	Patients with newly-diagnosed, limited-stage SCLC	Intervention: 3500 IU daily of bemiparin for a maximum of 26 weeks Control: No intervention	Survival outcome, thromboembolic events, major and minor bleeding, and thrombocytopenia	Open-label randomized controlled trial	3

SCLC=Small cell lung cancer, CT=Computed tomography, NSCLC=Nonsmall cell lung cancer

Table 2: Impact of heparin therapy on venous thromboembolism and safety outcomes in patients with lung cancer

Outcomes	Number of events		RR (95% CI)	I^2 (%)
	Heparin	Control		
Thromboembolic events	19/529	34/404	0.46 (0.27-0.80)	0
Bleeding	42/534	25/413	1.53 (0.96-2.45)	6.2
Major bleeding	12/492	7/371	1.43 (0.59-3.45)	0
Thrombocytopenia	78/335	95/332	0.86 (0.66-1.12)	14.5

RR=Relative risk, CI=Confidence interval

and suggested a higher benefit in patients with limited SCLC.[24] In the two RCTs, Lebeau *et al*. explored the effect of UFH[13] while Altinbas *et al*. combined LMWH with chemotherapy,[14] and they both observed a discernible effect on overall survival. Our current meta-analysis has included two new studies compared with Akl's meta-analysis. The ABEL study enrolled patients with limited-stage SCLC and reported a significant overall survival advantage in the group assigned to LMWH, although the trial included only 38 patients.[15] Whereas the study by van Doormaal did not show a survival benefit of nadroparin in patients with NSCLC.[16] Zhang *et al*. also conducted a systematic review addressing the same question as our review.[25] They calculated pooled relative risk (RRs) at 6 months, 1 year, and 2 years, whereas we calculated pooled HRs, which represents instantaneous rate and may be more persuasive. This study has shown that patients with limited-stage SCLC may benefit most from the heparin therapy.

Factors that may impact survival analysis include the prognosis of the patients and stage of the disease. In the FAMOUS and MALT studies which aimed to evaluate the impact of LMWH on cancer survival, the subgroup of patients with a prior life expectancy experienced a superior survival benefit.[26,27] We also confirmed that limited-stage SCLC patients gain a superior survival advantage from heparin therapy than patients with extensive SCLC. Besides, whether the patients with cancer can benefit from heparin therapy depends on tumor types since VTE risk varies among different histological types of cancer.[28-30] The highest risk of VTE was observed among patients with cancers of the pancreas, brain, and lung[31] as a result the decrease in VTE is most apparent among pancreatic and lung cancer.[32,33] Thus, VTE prophylaxis may be preferred in these specific populations. In addition, the optimal dose, the type, and the duration of the treatment with LMWH or UFH remain inconclusive. For example, a recent retrospective study demonstrated that LMWH usage for more than 4 months increased risk for OS and PFS significantly in SCLC cases with comparison to LMWH usage for <4 months while there was not any significant difference between the duration of 4–6 months and more than 6 months.[34] Thus, further studies aimed at additional heparin therapy to standard cancer treatment in lung cancer patients are warranted.

It is convincingly demonstrated that the survival benefit might at least in part attribute to a decrease in venous thromboembolic events. However, there is a substantial body of evidence showing that heparin has advantages beyond just the prevention and treatment of VTE. We speculate that the antitumor effects, anti-angiogenic effects, and immunomodulatory effects of heparin may also deliver survival benefit.[35-37]

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The major concern for using heparin is its potential bleeding risk. Other potential disadvantages, including health-care costs and patient discomfort from daily subcutaneous injections, should also be valued. Based on the results of the current meta-analysis, LMWH increased the risk of major bleeding[38] while some studies reported no apparent increase in major bleeding.[32,39] Although both bleeding and major bleeding complications were not significantly increased by the heparin treatment in our meta-analysis, bleeding complication, especially major bleeding, must be handled cautiously owing to its lethality. Thence, further studies are needed to define risk stratification of VTE, bleeding, and other complications to highlight which patients should receive prophylaxis. Cost-effectiveness analysis should be appraised as well since most of the studies did not report hospitalization rate and cost.

Several limitations of our study should be noted. First of all, it is still not known whether the addition of heparin has similar effects in different lung cancer type due to a rather small number of studies. In our meta-analysis, only one included study was eligible for the analysis of NSCLC. Hence, the data were insufficient to reveal a statistically significant potential benefit. Further powered studies are required. Second, we cannot rule out that different dose, duration, and type of heparin may have different effects as no adequate information was available. Third, the number of patients included in the analyzed studies was relatively small, so the inclusion of future trials might change the results. These limits decreased powerfulness of our results.

CONCLUSION

Our systematic review and meta-analysis reveal that administration of heparin (mainly LMWH) as primary thromboprophylaxis for lung cancer patients without indication for anticoagulants is associated with apparent survival benefit, particularly in limited-stage SCLC. Thromboembolic events were reduced by the intervention of heparin while the risk for bleeding and major bleeding was not significantly increased and neither the rate of thrombocytopenia.

Financial support and sponsorship

Nil.

Conflicts of interest

There are no conflicts of interest.

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